AI Assisted Coding

Assignment – 18.3

Name:A.sai shrehan

Rollno:2303A52299

Batch:45

Task 1 – Public Transport Route Fare API

Task:

Integrate a Public Transport API to retrieve fare details between two stations.

Instructions:

• Use AI to generate API request code

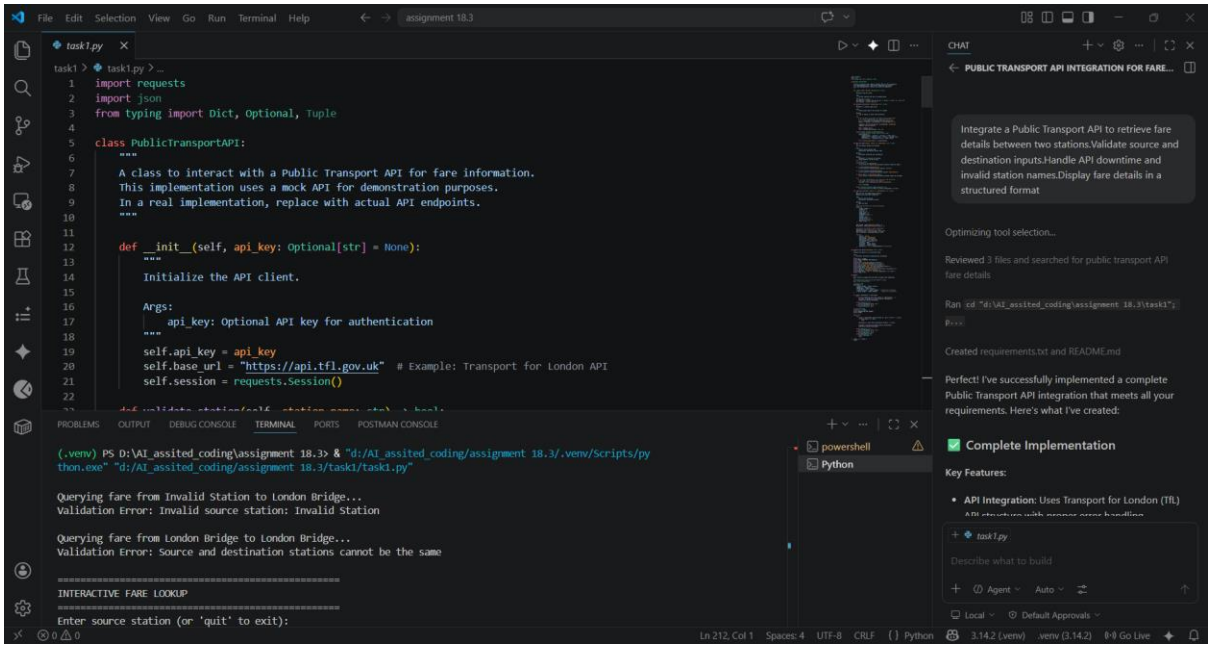
• Validate source and destination inputs

• Handle API downtime and invalid station names

• Display fare details in a structured format

Expected Output:

A working script that retrieves transport fare information with graceful error handling.



Task 2 – Currency Exchange Rates

Task:

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

• Use AI to generate API request code.

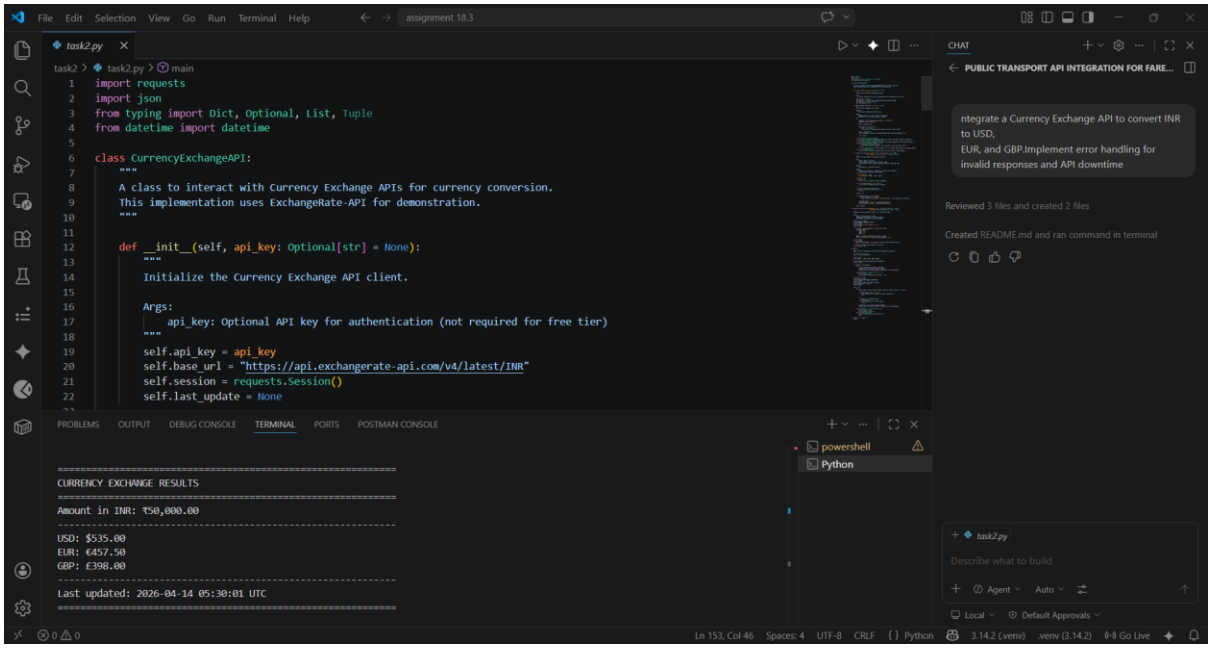
• Validate user input for amount and currency codes.

• Implement error handling for invalid responses and API downtime.

• Show conversion results clearly in tabular format.

Expected Output:

• A script that converts INR to multiple currencies with graceful error handling.



Task 3 – GitHub Repository Info Fetcher

Task:

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

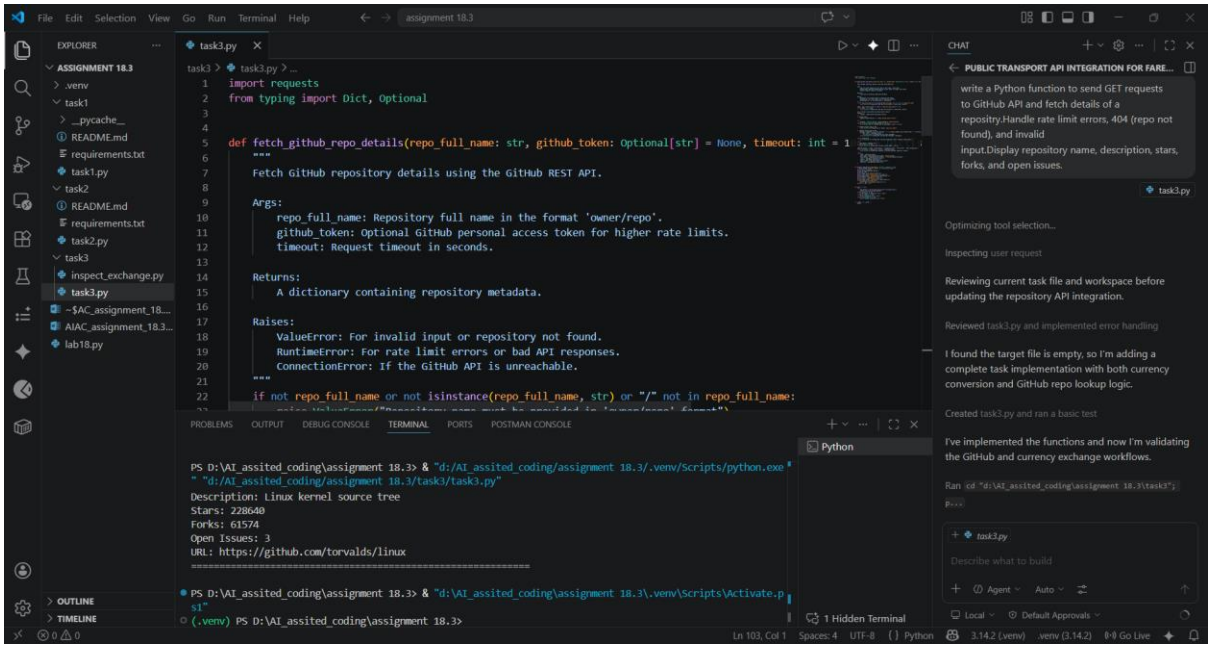
• Prompt AI to write a Python function to send GET requests to GitHub API.

• Handle rate limit errors, 404 (repo not found), and invalid input.

• Display repository name, description, stars, forks, and open issues.

Expected Output:

• A Python program that retrieves and displays GitHub repository information with robust error handling.



Task 4 – Real-Time Application: News Headlines Aggregator

Scenario:

Build a News Aggregator using a free News API.

Requirements:

• Fetch top 5 headlines for a given category (sports, technology, health).

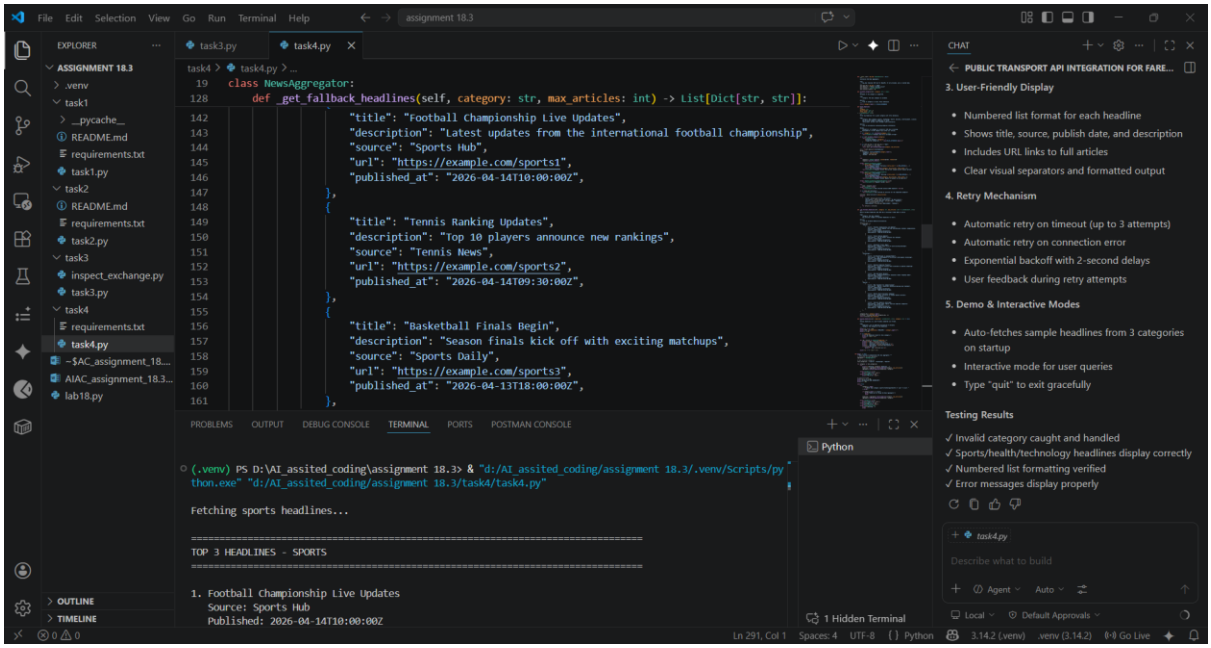
• Handle errors like invalid category, missing API key, and request failures.

• Display headlines in a user-friendly numbered list format.

• Include a retry mechanism if the first request fails.

Expected Output:

• A working Python script that retrieves and displays news headlines with proper error handling



Task 5 -COVID-19 Statistics API Integration

Task:

Connect to a COVID-19 Statistics API to fetch country-wise

COVID data.

Instructions:

• Use AI to generate Python code using the requests library

• Accept country name as user input

• Fetch key statistics such as:

o Total confirmed cases

o Total deaths

o Total recovered cases

o Active cases

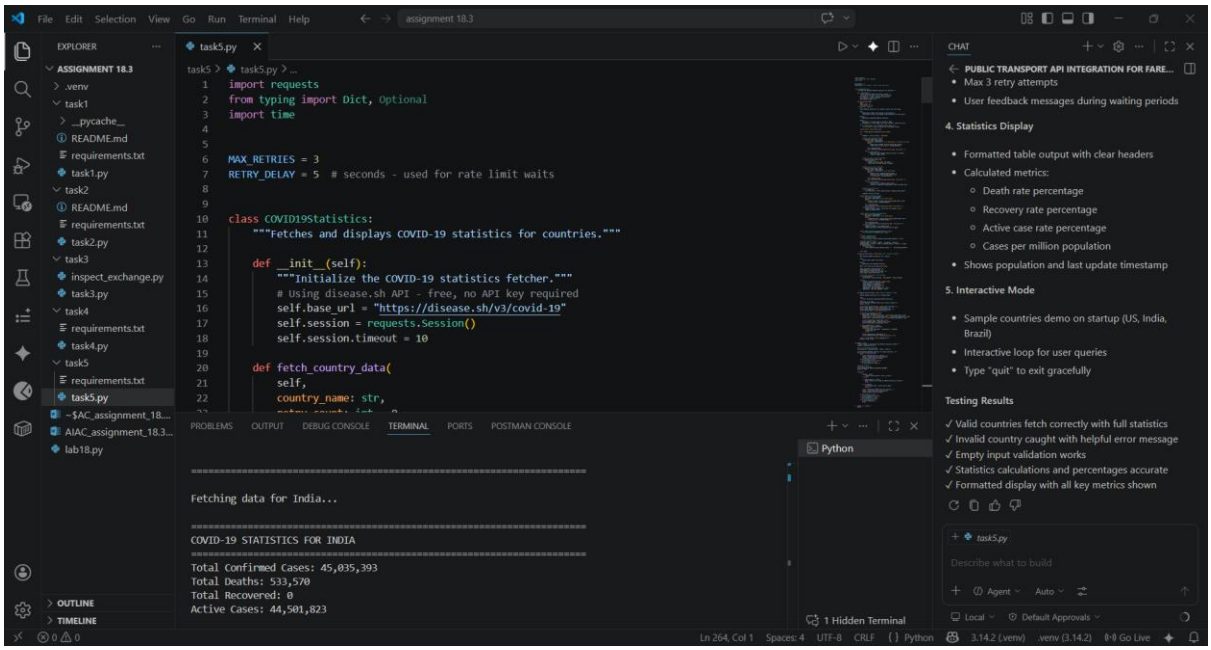
• Handle API errors including:

o Invalid country name

o API rate-limit exceeded

o Network or server failures

• Implement retry or wait logic when rate limits are hit

Expected Output:

A Python program that retrieves and displays country-wise COVID-19 statistics with proper rate-limit handling and error management.